Systems of Linear Equations (AA HL)

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Augmented Matrices and Gaussian Elimination**

Solve the first-order linear differential equation $\frac{dy}{dx} + \frac{2}{x}y = 4x$ for $x > 0$, given the initial condition $y(1) = 2$.

1. Identify $P(x) = \frac{2}{x}$ and $Q(x) = 4x$. 2. Calculate the integrating factor $\mu(x)$: $\mu(x) = \exp\left(\int \frac{2}{x}\,dx\right) = e^{2\ln x} = e^{\ln(x^2)} = x^2$. 3. Multiply through by x^2: $x^2\frac{dy}{dx} + 2xy = 4x^3 \Rightarrow \frac{d}{dx}[x^2y] = 4x^3$. 4. Integrate both sides: $x^2y = \int 4x^3\,dx = x^4 + C$. 5. Apply $y(1) = 2$: $(1)^2(2) = 1^4 + C \Rightarrow 2 = 1 + C \Rightarrow C = 1$. 6. Explicit solution: $y = \frac{x^4+1}{x^2} = x^2 + \frac{1}{x^2}$.

2 **[Foundational] • Target: Geometric Interpretation of 3 x 3 Systems**

Two independent random variables are normally distributed as $X \sim N(50, 16)$ and $Y \sim N(45, 9)$. Find the probability that $X > Y$.

1. Define the difference variable $W = X - Y$. The condition $X > Y$ is equivalent to $W > 0$. 2. Since X and Y are independent normals, W is normally distributed: Mean: $\mu_W = E[X] - E[Y] = 50 - 45 = 5$. Variance: $\sigma_W^2 = \text{Var}(X) + \text{Var}(Y) = 16 + 9 = 25 \Rightarrow \sigma_W = 5$. Thus $W \sim N(5, 25)$. 3. Compute $P(W > 0)$: $z = \frac{0-5}{5} = -1.0$. $P(W > 0) = P(Z > -1.0) = P(Z < 1.0) = \Phi(1.0) \approx 0.8413 \approx 0.841$.

3 **[Foundational] • Target: Determinants and Cramer's Rule**

Solve the second-order homogeneous linear differential equation $y'' - 5y' + 6y = 0$ satisfying initial conditions $y(0) = 1$ and $y'(0) = 3$.

1. Write the characteristic equation: $m^2 - 5m + 6 = 0 \Rightarrow (m-2)(m-3) = 0 \Rightarrow m_1 = 2,\ m_2 = 3$. 2. General solution: $y(x) = Ae^{2x} + Be^{3x}$. 3. Derivative: $y'(x) = 2Ae^{2x} + 3Be^{3x}$. 4. Apply initial conditions: $y(0) = 1 \Rightarrow A + B = 1$. $y'(0) = 3 \Rightarrow 2A + 3B = 3$. Multiply first equation by 2: $2A + 2B = 2$. Subtract from second: $B = 1$. Then $A = 1 - 1 = 0$. Particular Solution: $y(x) = e^{3x}$.

4 **[Foundational] • Target: Parameterized Solutions and Free Variables**

Consider the points $A(1, -2, 3)$, $B(4, 0, 5)$, and $C(p, 1, 2)$. If the vectors $\vec{AB}$ and $\vec{AC}$ are perpendicular, find the value of p.

First, find the vector $\vec{AB}$: $\vec{AB} = B - A = (4-1, 0-(-2), 5-3) = (3, 2, 2)$. Next, find the vector $\vec{AC}$: $\vec{AC} = C - A = (p-1, 1-(-2), 2-3) = (p-1, 3, -1)$. Two vectors are perpendicular if their dot product is zero. Therefore, $\vec{AB} \cdot \vec{AC} = 0$. $(3)(p-1) + (2)(3) + (2)(-1) = 0$ $3p - 3 + 6 - 2 = 0$ $3p + 1 = 0$ $3p = -1$ $p = -\frac{1}{3}$ Final Answer: $p = 4$

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5 [Foundational] • Target: Augmented Matrices and Gaussian Elimination

A fair six-sided die is rolled. What is the probability of rolling a number greater than 4?

The possible outcomes when rolling a fair six-sided die are {1, 2, 3, 4, 5, 6}. There are 6 equally likely outcomes in total. The outcomes that are greater than 4 are {5, 6}. There are 2 favorable outcomes. The probability of an event is the ratio of the number of favorable outcomes to the total number of possible outcomes. Therefore, the probability of rolling a number greater than 4 is $\frac{2}{6} = \frac{1}{3}$. Final Answer: 1/3

6 [Foundational] • Target: Geometric Interpretation of 3 x 3 Systems

Phone calls arrive at an emergency center according to a Poisson process with an average rate of 3 calls per minute. Find: (a) the probability that at least 2 calls arrive in a 1-minute period; (b) the probability that exactly 5 calls arrive in a 2-minute period.

(a) For 1 minute, $X \sim \text{Po}(3)$: $P(X \geq 2) = 1 - P(X = 0) - P(X = 1) = 1 - e^{-3}\frac{3^0}{0!} - e^{-3}\frac{3^1}{1!} = 1 - e^{-3}(1 + 3) = 1 - 4e^{-3} \approx 1 - 0.19915 = 0.8009 \approx 0.801.$ (b) For 2 minutes, the rate scales linearly: $\lambda = 3 \times 2 = 6$, so $Y \sim \text{Po}(6)$: $P(Y = 5) = \frac{e^{-6} \cdot 6^5}{5!} = \frac{e^{-6} \cdot 7776}{120} = 64.8e^{-6} \approx 0.1606 \approx 0.161.$

7 [Foundational] • Target: Determinants and Cramer's Rule

A discrete random variable X takes values $0, 3, 4$ with probabilities $\frac{1}{10}, \frac{3}{10}, \frac{3}{5}$ respectively. Find $E(X)$.

Working through this statistics step: $E(X) = \sum xP(X = x) = 0(\frac{1}{10}) + 3(\frac{3}{10}) + 4(\frac{3}{5}) = \frac{33}{10}$ Final Answer: $\frac{33}{10}$ Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Parameterized Solutions and Free Variables

The region bounded by $y = \ln x$, the line $y = 0$, the line $y = 1$, and the y-axis is rotated 2π radians about the y-axis. Find the exact volume of the resulting solid.

1. Express x as a function of y: $y = \ln x \Rightarrow x = e^y$. 2. Apply the volume of revolution about the y-axis formula: $V = \pi\int_c^d [x(y)]^2\,dy = \pi\int_0^1 (e^y)^2\,dy = \pi\int_0^1 e^{2y}\,dy$. 3. Integrate: $V = \pi\left[\frac{e^{2y}}{2}\right]_0^1 = \frac{\pi}{2}(e^2 - e^0) = \frac{\pi(e^2 - 1)}{2}$.

9 [Intermediate] • Target: Augmented Matrices and Gaussian Elimination

Consider the function $f(x) = \frac{1}{2}x^2 - 3x + 5$. The graph of $y = f(x)$ is a parabola. Find the x-coordinate of the vertex of the parabola.

The x-coordinate of the vertex of a parabola given by the equation $f(x) = ax^2 + bx + c$ is found using the formula $x = -\frac{b}{2a}$. In this case, $a = \frac{1}{2}$ and $b = -3$. Therefore, the x-coordinate of the vertex is $x = -\frac{-3}{2(\frac{1}{2})} = -\frac{-3}{1} = 3$.

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10 **[Intermediate]** • **Target: Geometric Interpretation of 3 x 3 Systems**

Find $\int xe^{2x}\,dx$ (in terms of x, with constant C).

Working through this calculus step: $u = x,\ dv = e^{2x}dx \Rightarrow \int xe^{2x}dx = \frac{x}{2}e^{2x} - \int \frac{1}{2}e^{2x}dx = \frac{x}{2}e^{2x} - \frac{1}{4}e^{2x} + C$ Final Answer: $\frac{x}{2}e^{2x} - \frac{1}{4}e^{2x} + C$

11 **[Intermediate]** • **Target: Determinants and Cramer's Rule**

Find the number of solutions of $\sin(2\theta) = \frac{1}{2}$ in the interval $0^\circ \le \theta < 360^\circ$.

Final Answer: 4. Working through this additional-topics step:

$2\theta = 30^\circ, 150^\circ, 390^\circ, 510^\circ$ (within $0^\circ \le 2\theta < 720^\circ$) $\Rightarrow \theta = 15^\circ, 75^\circ, 195^\circ, 255^\circ$: 4 solutions.

12 **[Intermediate]** • **Target: Parameterized Solutions and Free Variables**

How many solutions does $\sin\theta = \sin 30^\circ$ have in the interval $0^\circ \le \theta < 360^\circ$? Show your complete algebraic work.

Final Answer: 2. Working through this additional-topics step:

$\theta = 30^\circ$ or $\theta = 180^\circ - 30^\circ = 150^\circ$: 2 solutions in $[0^\circ, 360^\circ)$.

13 **[Intermediate]** • **Target: Augmented Matrices and Gaussian Elimination**

How many solutions does $\sin\theta = \sin 60^\circ$ have in the interval $0^\circ \le \theta < 360^\circ$? Show your complete algebraic work.

Final Answer: 2. Working through this additional-topics step:

$\theta = 60^\circ$ or $\theta = 180^\circ - 60^\circ = 120^\circ$: 2 solutions in $[0^\circ, 360^\circ)$.

14 **[Intermediate]** • **Target: Geometric Interpretation of 3 x 3 Systems**

How many solutions does $\sin\theta = \sin 45^\circ$ have in the interval $0^\circ \le \theta < 360^\circ$? Show your complete algebraic work.

Final Answer: 2. Working through this additional-topics step:

$\theta = 45^\circ$ or $\theta = 180^\circ - 45^\circ = 135^\circ$: 2 solutions in $[0^\circ, 360^\circ)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Determinants and Cramer's Rule** • **Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra**

For which value of m does the system $x + my = 3,\ 2x + 4y = 6$ have infinitely many solutions? Show your complete algebraic work.

Final Answer: 2. For infinitely many solutions the second equation must be exactly twice the first: $2/1 = 2$ and $4/m = 2 \Rightarrow m = 2$ (and $6 = 2 \times 3$, consistent).

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16 [Challenging] • Target: Parameterized Solutions and Free Variables • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

How many solutions does the system $x + y = 4,\ 2x + 2y = 8$ have? Show your complete algebraic work.

Final Answer: Infinitely many. The second equation is exactly twice the first, so both describe the same line — infinitely many solutions.

17 [Challenging] • Target: Augmented Matrices and Gaussian Elimination • Cross-Lesson: 1.04 Partial Fractions

An investment grows by $3\%, 1\%, -5\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

Final Answer: -0.39\%. Working through this additional-topics step:
$\bar{r} = \sqrt[3]{1.03 \times 1.01 \times 0.95} - 1 \approx -0.39\%$

18 [Challenging] • Target: Geometric Interpretation of 3 x 3 Systems • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

An investment grows by $-6\%, -10\%, 4\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

Final Answer: -4.18\%. Working through this additional-topics step:
$\bar{r} = \sqrt[3]{0.94 \times 0.90 \times 1.04} - 1 \approx -4.18\%$

19 [Challenging] • Target: Determinants and Cramer's Rule • Cross-Lesson: 3.01 3D Vectors & the Cross Product

An investment grows by $-3\%, 2\%, -8\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

Final Answer: -3.09\%. Working through this additional-topics step:
$\bar{r} = \sqrt[3]{0.97 \times 1.02 \times 0.92} - 1 \approx -3.09\%$

20 [Challenging] • Target: Parameterized Solutions and Free Variables • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

An investment grows by $1\%, -6\%, 1\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

Final Answer: -1.39\%. Working through this additional-topics step:
$\bar{r} = \sqrt[3]{1.01 \times 0.94 \times 1.01} - 1 \approx -1.39\%$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Augmented Matrices and Gaussian Elimination

In row-reducing an augmented matrix, which row operation is NOT permitted?

a. Squaring all elements in a row ✓

b. Swapping two rows

c. Multiplying a row by a non-zero scalar

d. Adding a multiple of one row to another row

The three valid elementary row operations are row swaps, multiplying a row by a non-zero constant, and adding a scalar multiple of a row to another row. Squaring entries alters the solution space.

22 [Foundational] • Target: Augmented Matrices and Gaussian Elimination

An augmented matrix in row echelon form has final row $[0, 0, 0 \mid 5]$. What does this indicate about the system?

a. The system is inconsistent (has no solutions) ✓

b. The system has infinitely many solutions

c. The system has a unique solution with $z = 5$

d. The system has trivial solutions only

The row represents the contradictory equation $0x + 0y + 0z = 5$ (i.e. $0 = 5$), which cannot be satisfied by any real values; hence the system is inconsistent.

23 [Foundational] • Target: Augmented Matrices and Gaussian Elimination

An augmented matrix in row echelon form has final row $[0, 0, 0 \mid 0]$ with 2 leading pivots across 3 variables. What does this signify?

a. The system has infinitely many solutions parameterized by 1 free variable ✓

b. The system has a unique solution

c. The system has no solutions

d. The determinant of the matrix is 1

With 3 variables and 2 leading pivot columns, there is $3 - 2 = 1$ free variable, which gives a line of infinitely many solutions.

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24 [Foundational] • Target: Geometric Interpretation of 3 x 3 Systems

What is the geometric configuration of three linear equations in three variables that possess a unique solution?

a. Three planes intersecting at a single unique point ✓

b. Three planes intersecting along a common line

c. Three parallel non-intersecting planes

d. Three planes forming a triangular prism

A unique solution (x_0, y_0, z_0) means all three planes share exactly one point of intersection.

25 [Foundational] • Target: Geometric Interpretation of 3 x 3 Systems

If three planes intersect along a single straight line, what type of system is this?

a. Consistent dependent system with infinitely many solutions ✓

b. Inconsistent system with no solutions

c. Consistent independent system with 1 solution

d. Homogeneous system with trivial solution only

Intersection along a line means infinitely many common points exist, parameterized by 1 free variable.

26 [Foundational] • Target: Geometric Interpretation of 3 x 3 Systems

Three planes form the faces of an open triangular prism with no common line or point. How is this system classified?

a. Inconsistent (no solutions) ✓

b. Infinitely many solutions

c. Unique solution

d. Singular dependent

Each pair of planes intersects in a line, but all three lines are parallel and distinct, meaning there is no point common to all three planes simultaneously.

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27 [Foundational] • Target: Determinants and Cramer's Rule

Evaluate the 2×2 determinant $\det[[4, -3], [2, 5]]$.

a. 26 ✓

b. 14

c. -26

d. 20

$\det = ad - bc = 4(5) - (-3)(2) = 20 - (-6) = 26$.

28 [Foundational] • Target: Determinants and Cramer's Rule

Under what condition does a 3×3 system $A\mathbf{x} = \mathbf{b}$ have a unique solution?

a. $\det(A) \neq 0$ ✓

b. $\det(A) = 0$

c. $\mathbf{b} = \mathbf{0}$

d. $\text{Tr}(A) > 0$

A square system $A\mathbf{x} = \mathbf{b}$ has a unique solution if and only if the coefficient matrix A is non-singular, meaning $\det(A) \neq 0$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Determinants and Cramer's Rule

If $\det(A) = 0$ for a system $A\mathbf{x} = \mathbf{b}$, what can be concluded?

a. **The system either has no solutions or infinitely many solutions** ✓

b. The system must have no solutions

c. The system must have infinitely many solutions

d. The system has exactly 2 solutions

When $\det(A) = 0$, A is singular; depending on the consistency of the constant vector $\mathbf{b}$, the system has either zero solutions or infinitely many solutions.

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30 [Intermediate] • Target: Parameterized Solutions and Free Variables

Solve the system: $x + y + z = 6$ and $2x - y + z = 3$. Parameterize the solution line using $z = \lambda$.

a. $x = 3 - \frac{2}{3}\lambda, y = 3 - \frac{1}{3}\lambda, z = \lambda$ ✓

b. $x = 3 + \lambda, y = 3 - \lambda, z = \lambda$

c. $x = 9 - 2\lambda, y = \lambda, z = \lambda$

d. $x = 1, y = 2, z = 3$

Add equations: $3x + 2z = 9 \Rightarrow 3x = 9 - 2\lambda \Rightarrow x = 3 - \frac{2}{3}\lambda$. Subtract from first: $y = 6 - x - z = 6 - (3 - \frac{2}{3}\lambda) - \lambda = 3 - \frac{1}{3}\lambda$.

31 [Intermediate] • Target: Parameterized Solutions and Free Variables

For what value of k does the system $x + 2y = 4$ and $3x + 6y = k$ have infinitely many solutions?

a. $k = 12$ ✓

b. $k = 4$

c. $k = 0$

d. $k = 6$

Multiply the first equation by 3: $3x + 6y = 12$. For infinitely many solutions, the equations must be identical, so $k = 12$.

32 [Intermediate] • Target: Parameterized Solutions and Free Variables

If the homogeneous system $A\mathbf{x} = \mathbf{0}$ has non-trivial solutions (solutions other than $\mathbf{x} = \mathbf{0}$), what must be true about $\det(A)$?

a. $\det(A) = 0$ ✓

b. $\det(A) > 0$

c. $\det(A) = 1$

d. $\det(A) < 0$

A homogeneous linear system has non-trivial solutions if and only if the coefficient matrix is singular, i.e., $\det(A) = 0$.

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33 [Intermediate] • Target: Augmented Matrices and Gaussian Elimination

An investment grows by $6\%,\ -4\%,\ -10\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

a. -4.89\%

b. -0.89\%

c. **-2.89\% ✓**

d. -1.39\%

Working through this additional-topics step: $\bar{r} = \sqrt[3]{1.06 \times 0.96 \times 0.90} - 1 \approx -2.89\%$

34 [Intermediate] • Target: Geometric Interpretation of 3 x 3 Systems

An investment grows by $3\%, 2\%, 18\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

a. 8.93\%

b. 5.93\%

c. 9.43\%

d. **7.43\% ✓**

Working through this additional-topics step: $\bar{r} = \sqrt[3]{1.03 \times 1.02 \times 1.18} - 1 \approx 7.43\%$

35 [Intermediate] • Target: Determinants and Cramer's Rule

An investment grows by $16\%, 4\%, -5\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

a. 7.65\%

b. 1.65\%

c. **4.65\% ✓**

d. 3.15\%

Working through this additional-topics step: $\bar{r} = \sqrt[3]{1.16 \times 1.04 \times 0.95} - 1 \approx 4.65\%$

36 [Intermediate] • Target: Parameterized Solutions and Free Variables

An investment grows by $13\%,\ -7\%, 6\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

a. 1.66\%

b. 6.66\%

c. 2.16\%

d. **3.66\% ✓**

Working through this additional-topics step: $\bar{r} = \sqrt[3]{1.13 \times 0.93 \times 1.06} - 1 \approx 3.66\%$

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37 [Intermediate] • Target: Augmented Matrices and Gaussian Elimination

An investment grows by $1\%, 12\%, 0\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

a. **4.19\% ✓**

b. 7.19\%

c. 5.69\%

d. 1.19\%

Working through this additional-topics step: $\bar{r} = \sqrt[3]{1.01 \times 1.12 \times 1.00} - 1 \approx 4.19\%$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Geometric Interpretation of 3 x 3 Systems • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

An investment grows by $6\%, -1\%, 8\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

a. 2.76\%

b. **4.26\% ✓**

c. 1.26\%

d. 2.26\%

Working through this additional-topics step: $\bar{r} = \sqrt[3]{1.06 \times 0.99 \times 1.08} - 1 \approx 4.26\%$

39 [Challenging] • Target: Determinants and Cramer's Rule • Cross-Lesson: 3.01 3D Vectors & the Cross Product

An investment grows by $14\%, 7\%, -3\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

a. 2.77\%

b. **5.77\% ✓**

c. 8.77\%

d. 4.27\%

Working through this additional-topics step: $\bar{r} = \sqrt[3]{1.14 \times 1.07 \times 0.97} - 1 \approx 5.77\%$

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40 [Challenging] • Target: Parameterized Solutions and Free Variables • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

An investment grows by $10\%, 5\%, 5\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

a. 5.14\%

b. 8.14\%

c. **6.64\% ✓**

d. 9.64\%

Working through this additional-topics step: $\bar{r} = \sqrt[3]{1.10 \times 1.05 \times 1.05} - 1 \approx 6.64\%$

41 [Challenging] • Target: Augmented Matrices and Gaussian Elimination • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

An investment grows by $0\%, 19\%, -3\%$ in three successive years. Find the average annual (geometric mean) growth rate, to 2 decimal places.

a. 6.40\%

b. **4.90\% ✓**

c. 6.90\%

d. 7.90\%

Working through this additional-topics step: $\bar{r} = \sqrt[3]{1.00 \times 1.19 \times 0.97} - 1 \approx 4.90\%$

42 [Challenging] • Target: Geometric Interpretation of 3 x 3 Systems • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Which value of k makes the system $x + 2y = 5,\ 2x + 4y = k$ have infinitely many solutions?

a. **10 ✓**

b. 5

c. 20

d. 15

The second equation must be exactly twice the first: $k = 2(5) = 10$.

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43 [Challenging] • Target: Determinants and Cramer's Rule • Cross-Lesson: 4.02 The Poisson Distribution

The graph of the function $f(x) = ax^2 - 6x + c$ is shown below. The graph has a vertex at the point (3, -9) and passes through the point (0, 0). Determine the values of a and c.

a. $a = 1, c = 0$

b. $a = 2, c = 0$ ✓

c. $a = 1, c = 3$

d. $a = 2, c = 3$

The function is given by $f(x) = ax^2 - 6x + c$. The vertex of a parabola in the form $f(x) = Ax^2 + Bx + C$ is located at $x = -\frac{B}{2A}$. In this case, $A = a$ and $B = -6$. Therefore, the x-coordinate of the vertex is $x = -\frac{-6}{2a} = \frac{6}{2a} = \frac{3}{a}$. We are given that the vertex is at (3, -9), so the x-coordinate of the vertex is 3. Equating the two expressions for the x-coordinate of the vertex: $\frac{3}{a} = 3$ Multiplying both sides by a gives $3 = 3a$, so $a = 1$. Now, we use the y-c.

44 [Challenging] • Target: Parameterized Solutions and Free Variables • Cross-Lesson: 5.01 Further Integration Techniques

The graph of the function $f(x) = \frac{x^2 - ax + 6}{x - 1}$ is shown below. The graph has a vertical asymptote at $x = 1$ and a point of discontinuity at $(2, k)$. Find the value of a.

a. **4** ✓

b. 5

c. 6

d. 7

The function is given by $f(x) = \frac{x^2 - ax + 6}{x - 1}$. $x = 1$ is a vertical asymptote, which means that $x - 1$ is not a factor of the numerator, and when $x = 1$, the numerator is non-zero. The point of discontinuity is at $(2, k)$. This implies that $x = 2$ is a root of the numerator, so $f(2) = 0$ or the expression is undefined at $x = 2$ if $x - 2$ were a factor of the denominator. Since the denominator is $x - 1$, which is not zero at $x = 2$, the function must be zero at $x = 2$ for there to be a removable discontinuity (a.

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45 [Challenging] • Target: Augmented Matrices and Gaussian Elimination • Cross-Lesson: 5.02 Differential Equations

The graph of the function $y = f(x)$ is shown below. The function is of the form $f(x) = ax^2 + bx + c$. The vertex of the parabola is at $(2, -4)$. The parabola passes through the points $(0, 0)$ and $(4, 0)$. Determine the value of a.

a. **-1** ✓

b. 1

c. -2

d. 2

The general form of a quadratic function is $f(x) = ax^2 + bx + c$. The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. In this case, the vertex is $(2, -4)$, so $h = 2$ and $k = -4$. Thus, the function can be written as $f(x) = a(x - 2)^2 - 4$. We are given that the parabola passes through the point $(0, 0)$. Substitute these coordinates into the vertex form equation: $0 = a(0 - 2)^2 - 4$ $0 = a(-2)^2 - 4$ $0 = 4a - 4$ $4a = 4$ $a = 1$. Let's verify with the other given point.

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